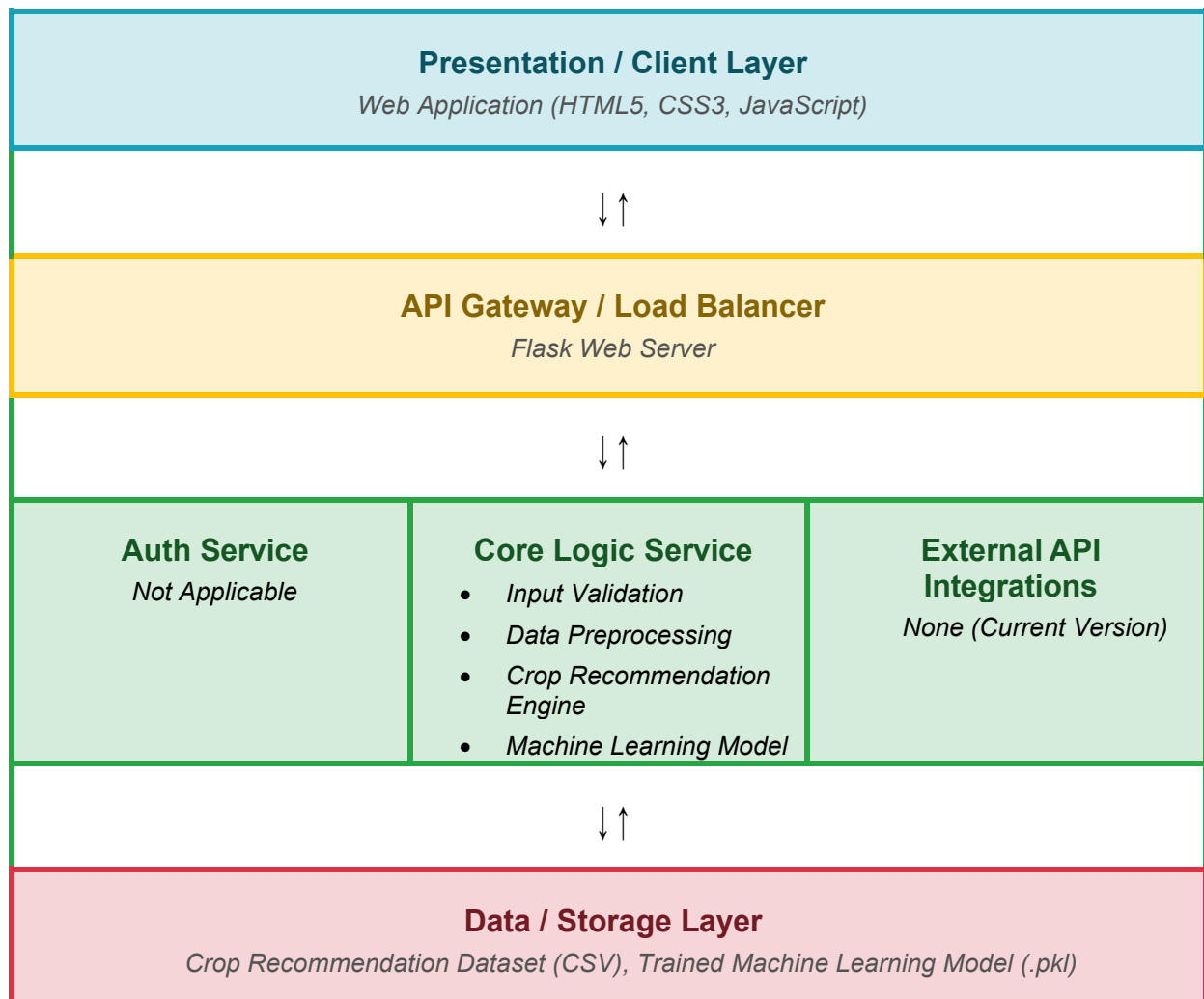


Solution Architecture

Date	6 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter soil and environmental parameters and view the recommended crop.	HTML5, CSS3, JavaScript
API Gateway (Flask Server)	Receives user requests, validates the input, and forwards the data to the machine learning model for prediction.	Flask, Python 3.10+
Microservices (Prediction Module)	Performs data preprocessing, loads the trained .pkl model, and predicts the most suitable crop based on user inputs.	Scikit-learn, Pandas, NumPy, Pickle
Database (Data Storage)	Stores the crop recommendation dataset and the trained machine learning model used for prediction.	CSV Dataset, Pickle (.pkl)